Create Graphs 10

Recall:

Data: *Data is a collection of information. This can be numbers, words, or descriptions of something.*

10. Create Graphs

Often, we have data that consists of a large set of numbers. We can create lists or tables, but there are more interesting ways to show data. If we show it in a visual manner, data becomes easier to read and interpret.

Graphs and **charts** are used to visualise data.

Recall:

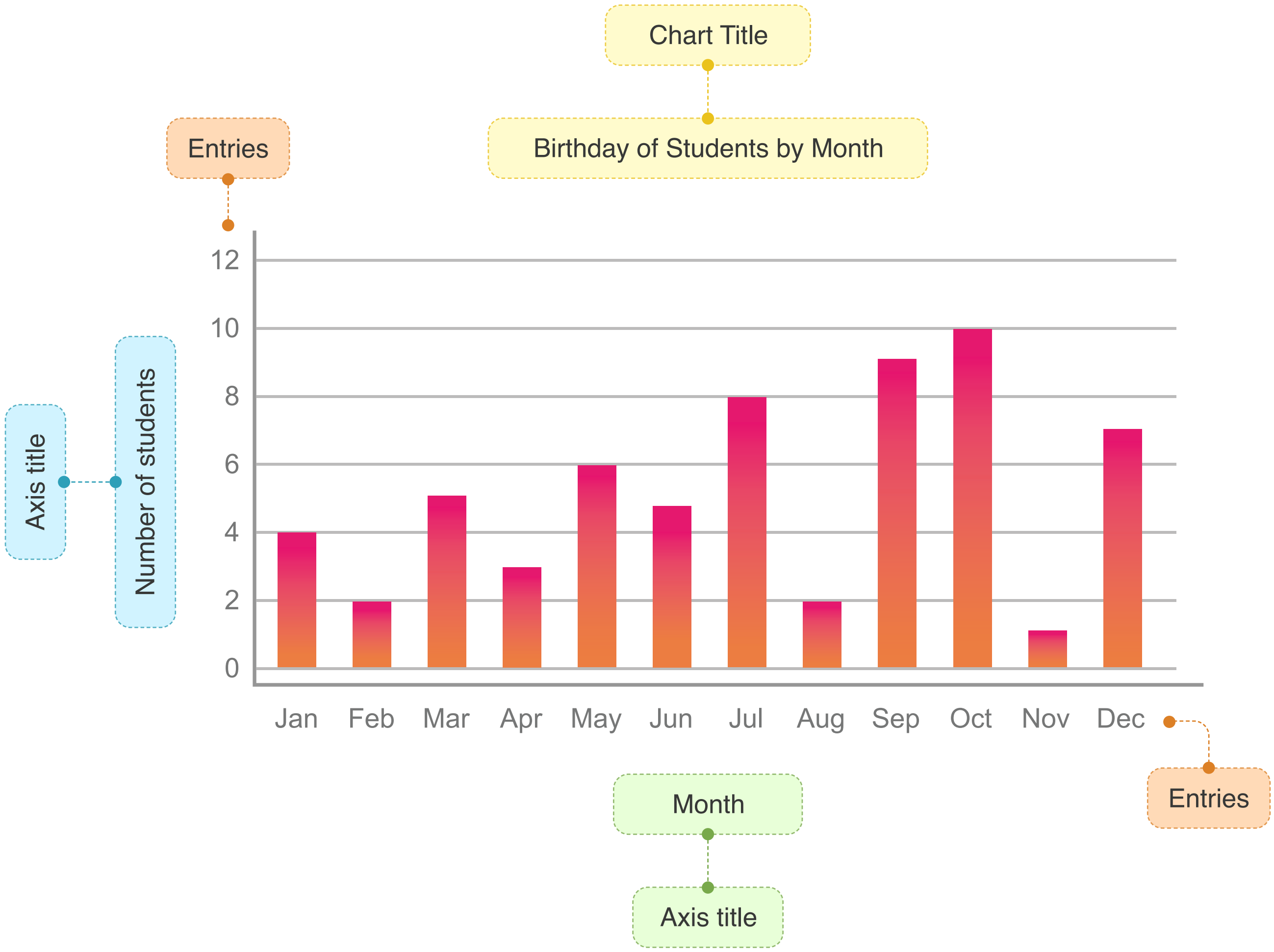
Pictograph: *A pictograph is a way to show data with pictures. It is easy to read as we can understand it when we look at it. A pictograph is a type of graph.*

What is a Graph?

- A graph is a pictorial representation that shows data with lines, bars, dots, shapes, or portions of a circle.
- It is a diagram that shows the relation between two or more things.
- For example:

Months	Students	Months	Students
January	𝍷𝍷𝍷𝍷	July	𝍸 𝍷𝍷𝍷
February	𝍷𝍷	August	𝍷𝍷
March	𝍸	September	𝍸 𝍷𝍷𝍷𝍷
April	𝍷𝍷𝍷	October	𝍸 𝍸
May	𝍸 𝍷	November	𝍷
June	𝍸	December	𝍸 𝍷𝍷

We have data of students and their birthdays in each month, and we want to find a pattern. It is easier to observe a graph than to read each entry separately.



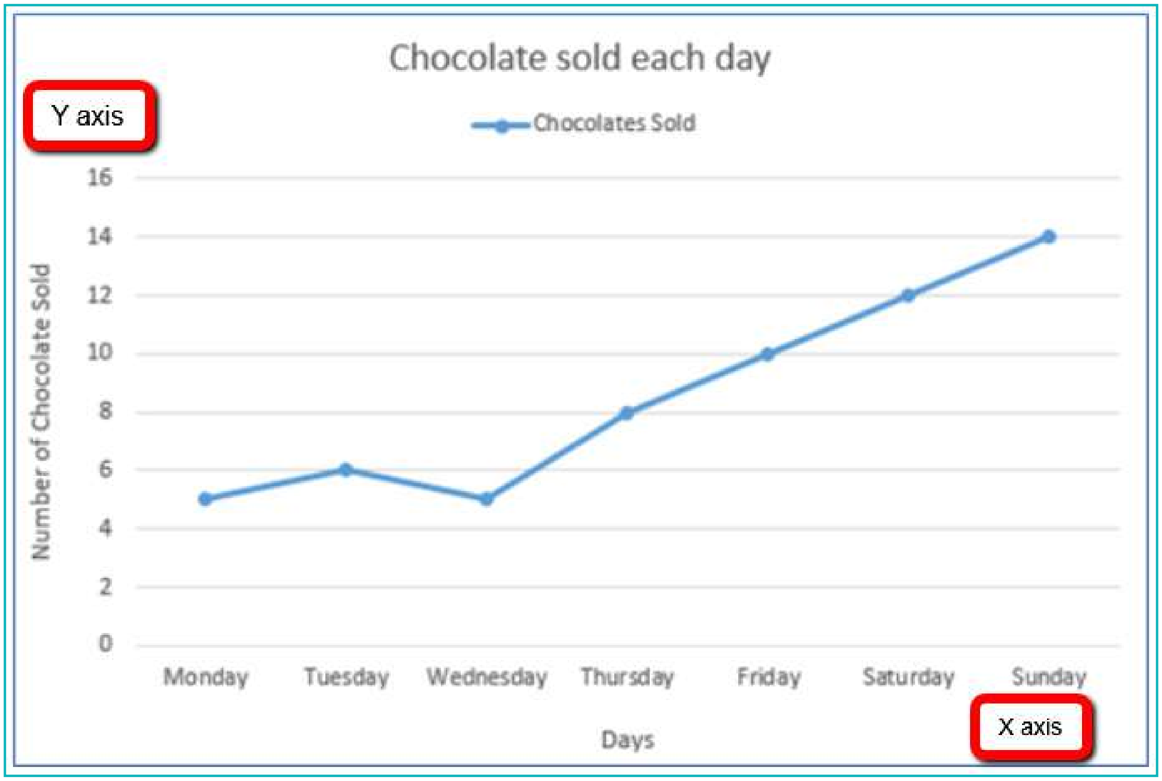
- To interpret a graph, read the **chart title, axis titles,** and look at the **entries.** Then study the graph to understand what it shows.
- When we use a graph to represent the data, bars are used to show the number of students who celebrate birthdays in that particular month. **The taller the bar, the higher is the number of students who celebrate their birthdays that month.**

Types of Graphs

a. Line Graph

A line graph is used to present data which changes over time.

For example: The data includes the number of chocolates sold each day in a week.

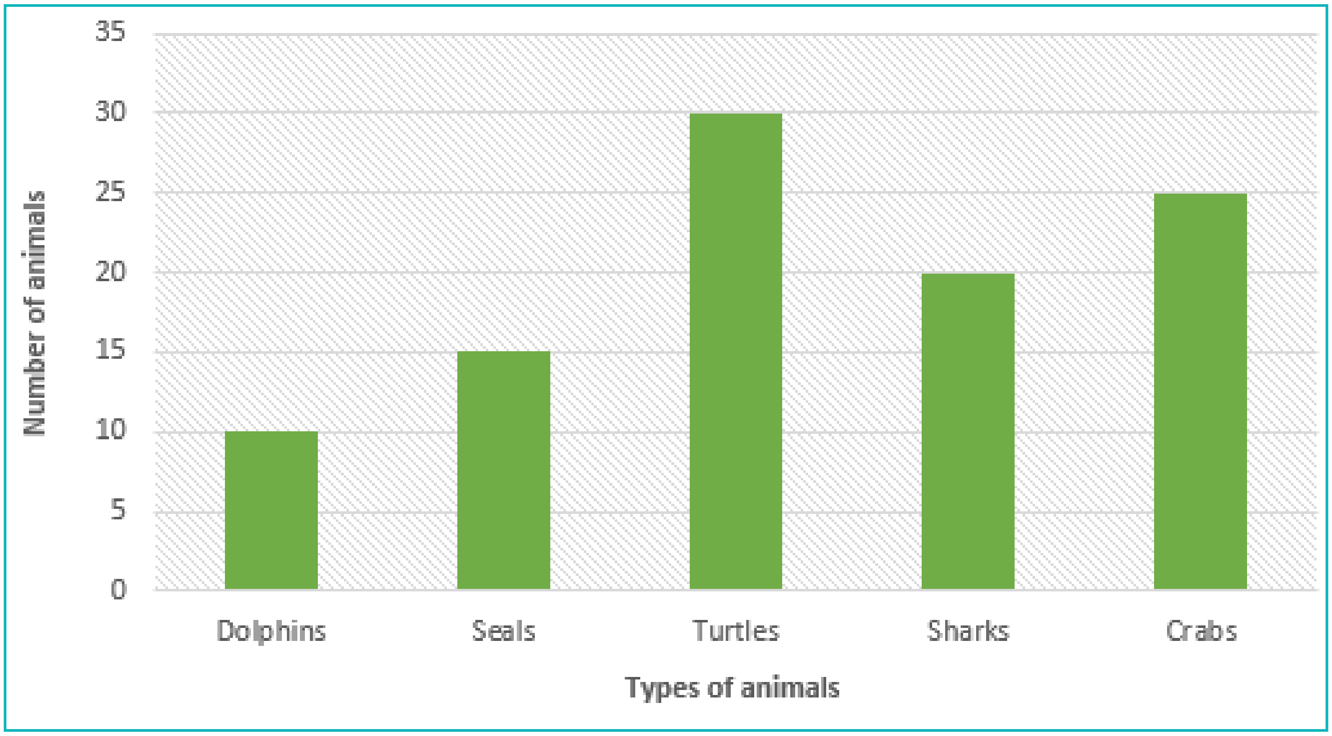


- In this graph, the **X-axis** (line from left to right) shows the **days of the week.** The **Y-axis** (line from top to bottom) shows the **number of chocolates sold.**
- Each dot shows the number of chocolates sold that day. Here, 5 chocolates were sold on Monday, 6 on Tuesday, 5 on Wednesday, 8 on Thursday, 10 on Friday, 12 on Saturday, and 14 on Sunday.
- A line is used to connect the plotted dots. The line shows the change in the number of chocolates sold over a week.

b. Bar Graph

A bar graph uses bars that represent numbers to shows data. The numbers are not dependent on each other.

For example: On an aquarium visit, the class saw 10 dolphins, 15 seals, 30 turtles, 20 sharks and 25 crabs. We can represent the types of animals and their numbers on a graph.

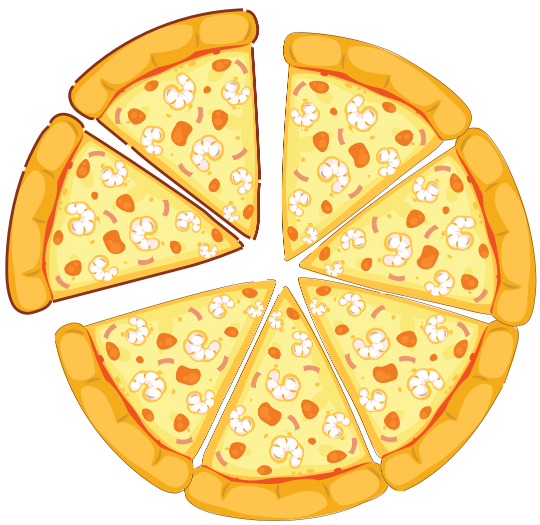


- Bars are used to show the numbers of each animal. **The taller the bar, the higher is the number of animals seen.**
- When we look at this graph, we can immediately tell that turtles were seen the most and dolphins were seen the least number of times.

c. Pie chart or Circle graph

A bar graph uses bars that represent numbers to shows data.
A pie chart uses "pie slices" to show portion of a part out of the total. A circle is split into sections, so a pie chart is also called a **circle graph.** Each section of a pie chart is called a sector.

For example: A pizza has 8 slices. Out of those, Yash ate 1 slice and Arya ate 1 slice. It means 2 out of 8 pizza slices were eaten and 6 out of 8 pizza slices still remains.



Each type of graph displays data in a different way. Some are better suited than others for different uses.

How do we choose which type of graph to use?

Line graph	Bar graph	Pie chart
A line graph is used to track changes over periods of time.	A bar graphs is mainly used to compare a quality between different groups.	A pie chart is used to compare parts of a whole. It does not show changes over time.
E.g.: Number of chocolates sold each day over one week.	E.g.: Number of students who celebrate their birthdays in each month.	E.g.: A pizza has 8 slices. Out of those, Yash ate 1 slice and Arya ate 1 slice. It means 2 out of 8 pizza slices were eaten and 6 out of 8 pizza slices still remains.

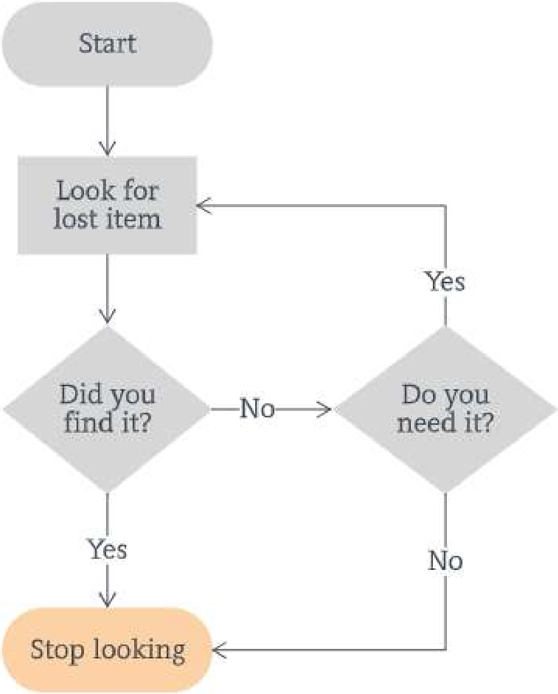
A graph is a type of chart, but it is not the only type of chart.

What is a chart?

- A chart represents data in the form of a table, graph, or diagram.
- The data is represented by symbols like lines, bars, or the sectors in a pie chart.
- All graphs are charts, but not all charts are graphs.

For example:

a. Flowchart: A flowchart is a diagram that represents a process or a step-by-step approach to solve a task.



b. Table: A table uses rows, columns, and cells to represent data.

Favourite Colour	Boys	Girls
Red	3	2
Green	5	2
Blue	6	6
Yellow	4	3
Pink	3	5
Purple	2	4

ICT 360

Exercise

Fill in the blanks

1. ____________________ and ____________________ are used to visualise data.

2. A pie chart is also known as ____________________.

State whether True or False

3. All charts are graphs.

▲ ____________________

4. A graph is a diagram which represents the relation between two or more things.

▲ ____________________

Answer the following question

5. What is a pie chart?

▲ ____

____

Activity 1

The list below shows the temperature (in Celsius) of New Delhi recorded for 10 days in the month of March. Let's create a graph to observe the change in temperature.

1. July 2023: 34 C
2. July 2023: 33 C
3. July 2023: 36 C
4. July 2023: 36 C
5. July 2023: 38 C
6. July 2023: 35 C
7. July 2023: 32 C
8. July 2023: 35 C
9. July 2023: 36 C
10. July 2023: 37 C

Before we plot a graph, we will add all the data in a spreadsheet.

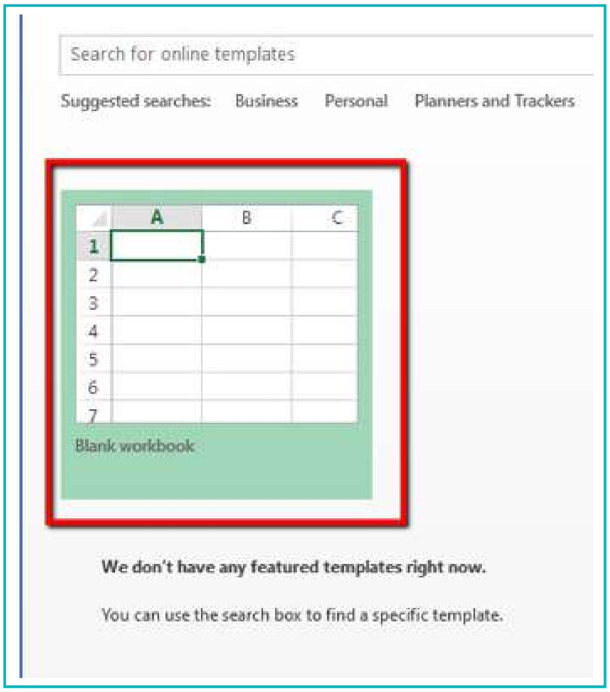
Recall:

Spreadsheet: *It is a set of rows and columns where the data can be stored.*

Steps

1. Search for Excel in the taskbar and launch the software.

2. To create a new spreadsheet, click on **blank workbook.**



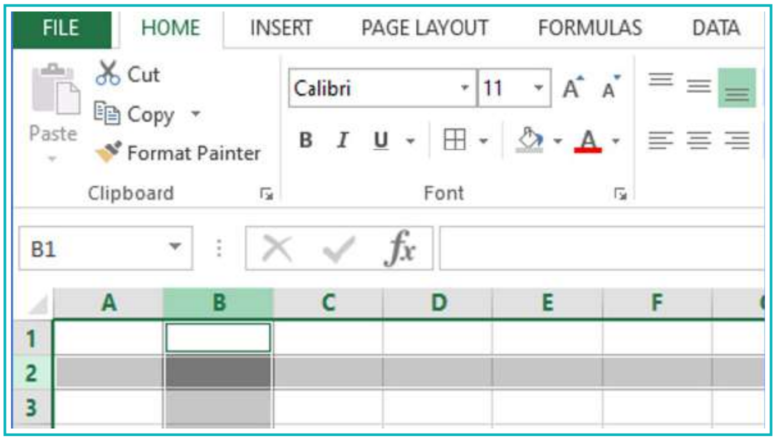
3. Start to enter data in any cell in the sheet. **For example:** Here we will use **B** column and **2nd** row

Recall:

Column: *Columns are the cells arranged vertically (top to bottom).*

Row: *Rows are the cells arranged horizontally (left to right).*

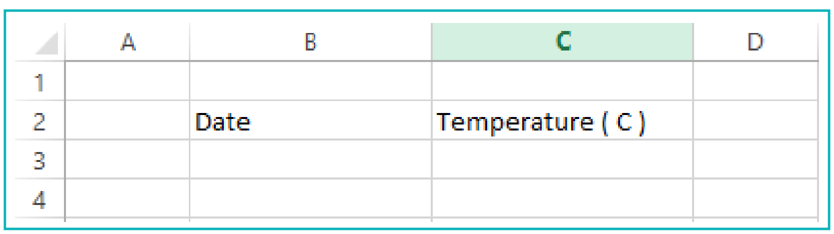
Steps



4 Generally, columns are used to enter the **type of data.** In this case, the types of data are:

a. Date: In this column, we will enter the dates when the temperature was recorded.

b. Temperature (C): In this column, we will write the temperature on that day.



Note:

Add a space between C and () e.g. (C). If there is no space between them, it converts to a different symbol in excel.

Recall:

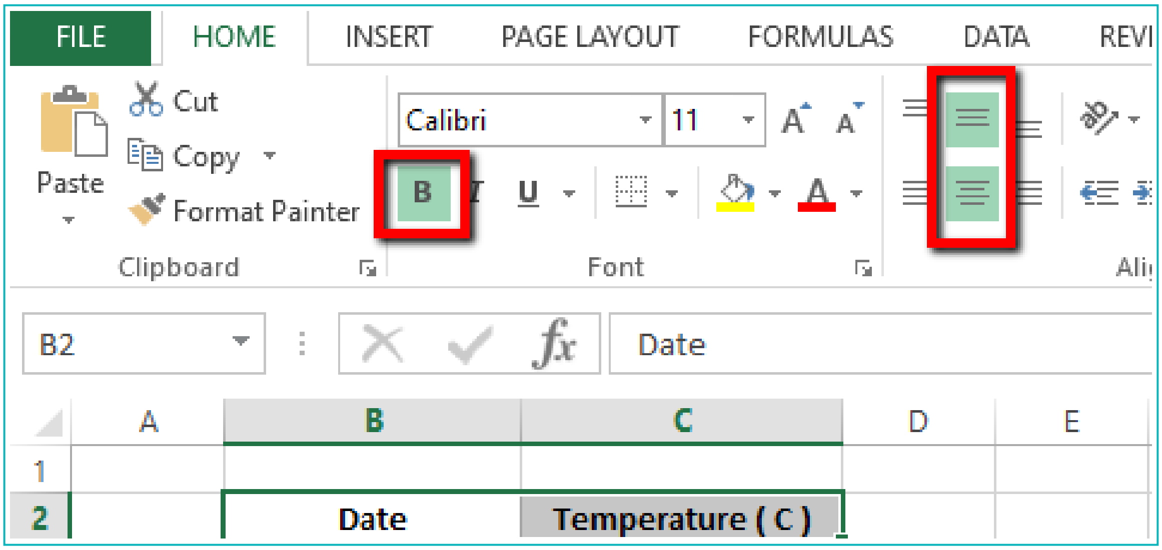
If the cell is too small to fit the data, we can change its size. Drag the border of the row or column to change the width or height.

Steps

5. Make **columns B** and **C** bigger to fit the data.

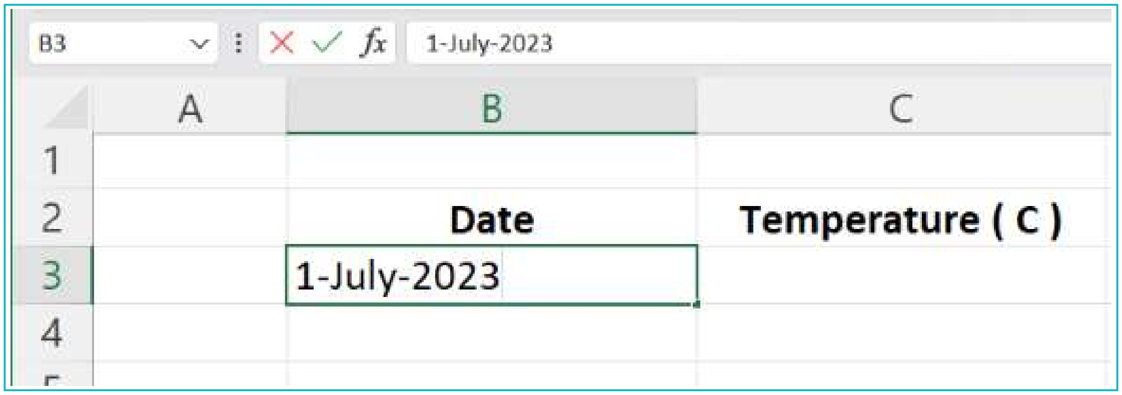
	A	B	C	D
1				
2		Date	Temperature (C)	
3				
4				
5				

6. To highlight the headings, use **Bold** and **Center.**

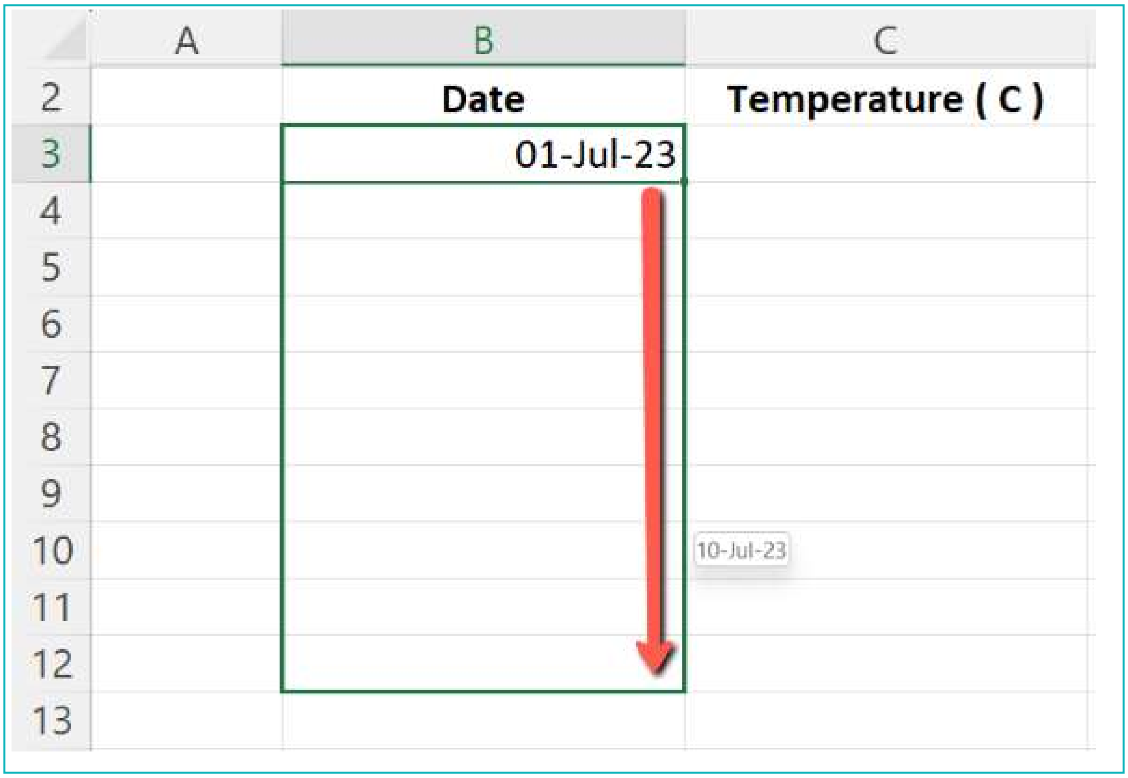


Steps

7 Rows are used for entries. In this case, the entries are the dates and temperatures on each date. Enter the 1st date as shown:



8 Select the date cell and drag down the **Fill Handle** (bottom right corner) till it reaches the final date.



Steps

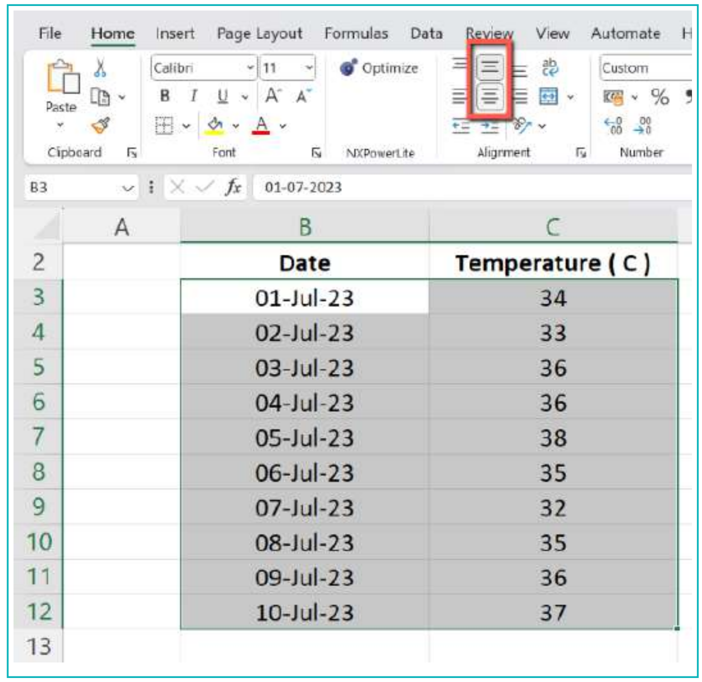
	A	B	C
2		Date	Temperature (C)
3		01-Jul-23	
4		02-Jul-23	
5		03-Jul-23	
6		04-Jul-23	
7		05-Jul-23	
8		06-Jul-23	
9		07-Jul-23	
10		08-Jul-23	
11		09-Jul-23	
12		10-Jul-23	
13			

9 Enter the temperatures for each date.

	A	B	C
2		Date	Temperature (C)
3		01-Jul-23	34
4		02-Jul-23	33
5		03-Jul-23	36
6		04-Jul-23	36
7		05-Jul-23	38
8		06-Jul-23	35
9		07-Jul-23	32
10		08-Jul-23	35
11		09-Jul-23	36
12		10-Jul-23	37
13			

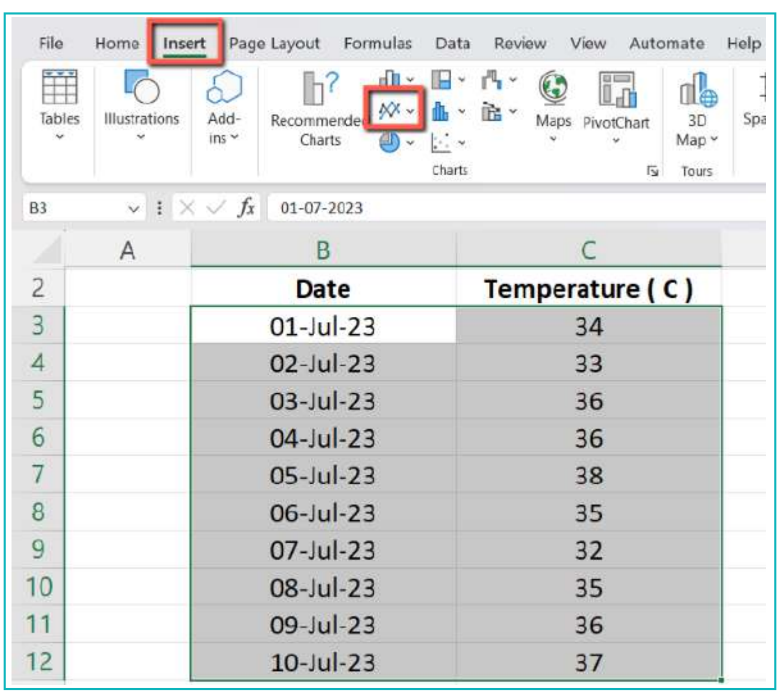
Steps

10 Right now, all the entries are aligned to the right edge of the cells. To make it look uniform, select all the data and use **centre alignment** to format it.



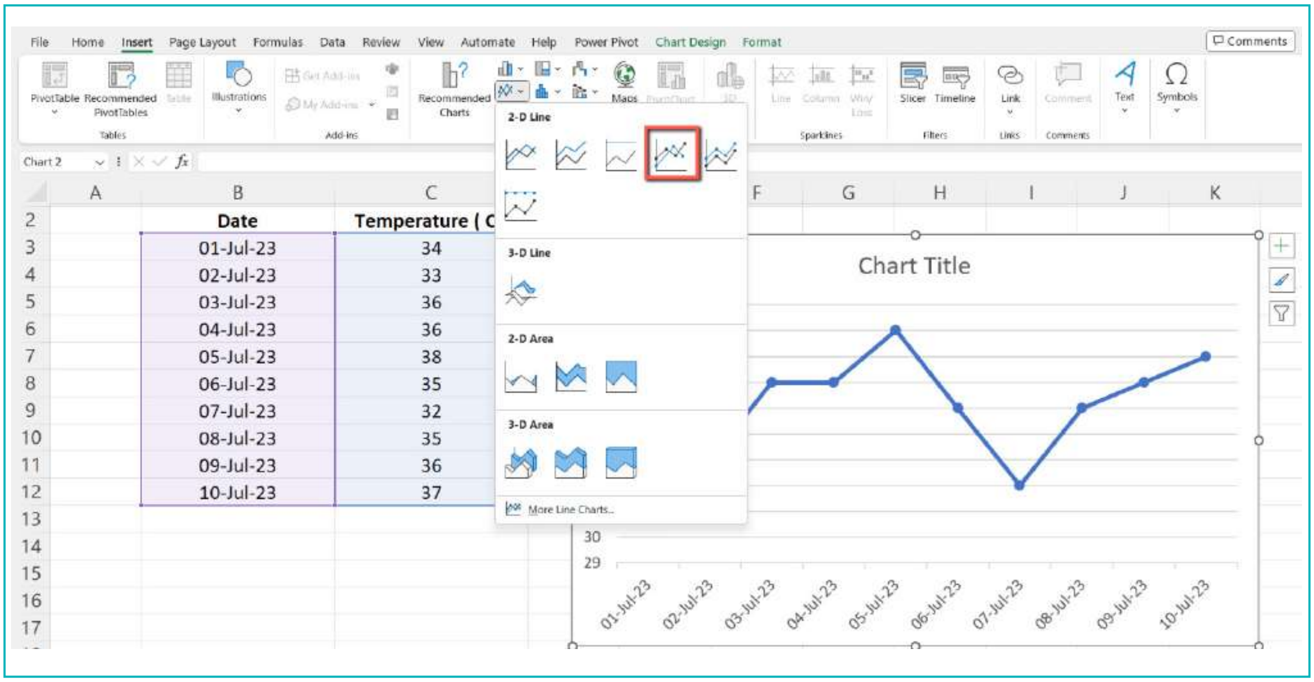
We want to view the **change in temperature** over a period of 10 days. For this, we will use a line graph to display the data.

11 To display the data in a line graph, select the data and click on the **Insert** tab. Under charts, select **Insert Line Graph.**



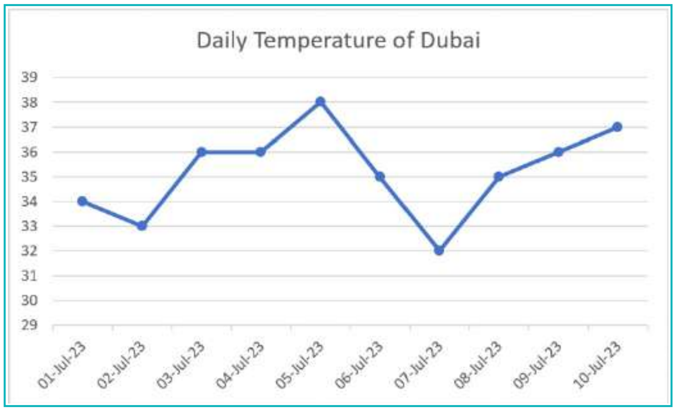
Steps

12 Select the **Line with markers** to create a line graph of the data.



13 Add a chart title: Double click on **chart title** and change it to **Daily temperature of Dubai.**

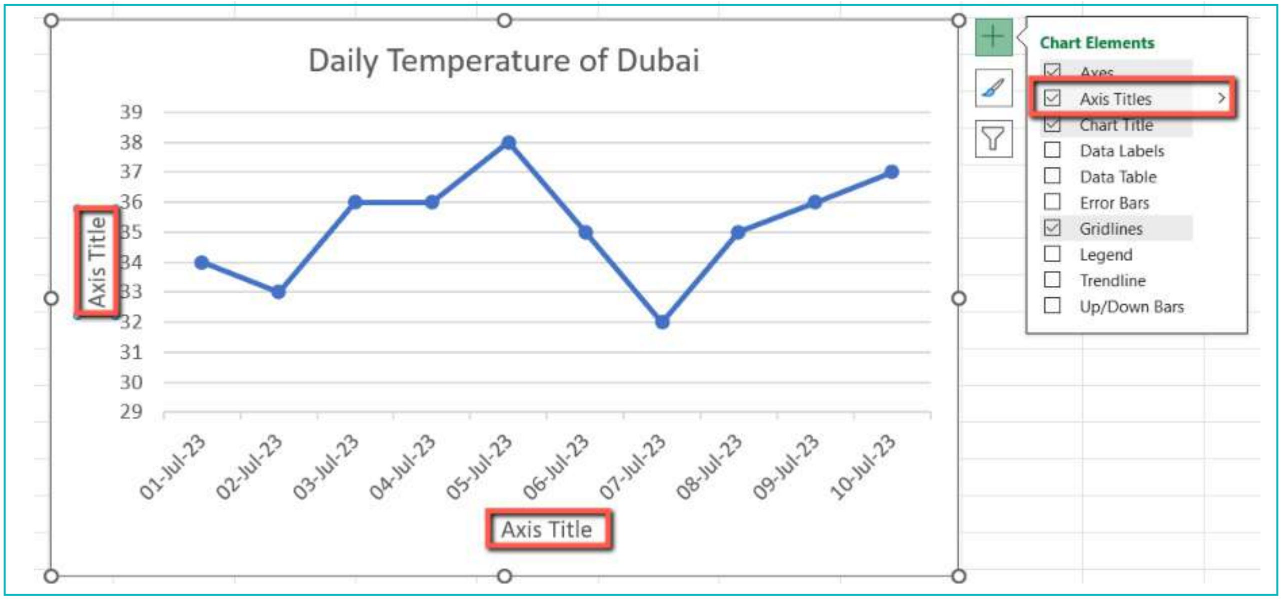
A chart title will tell you what the graph is about. It should be short and to the point.



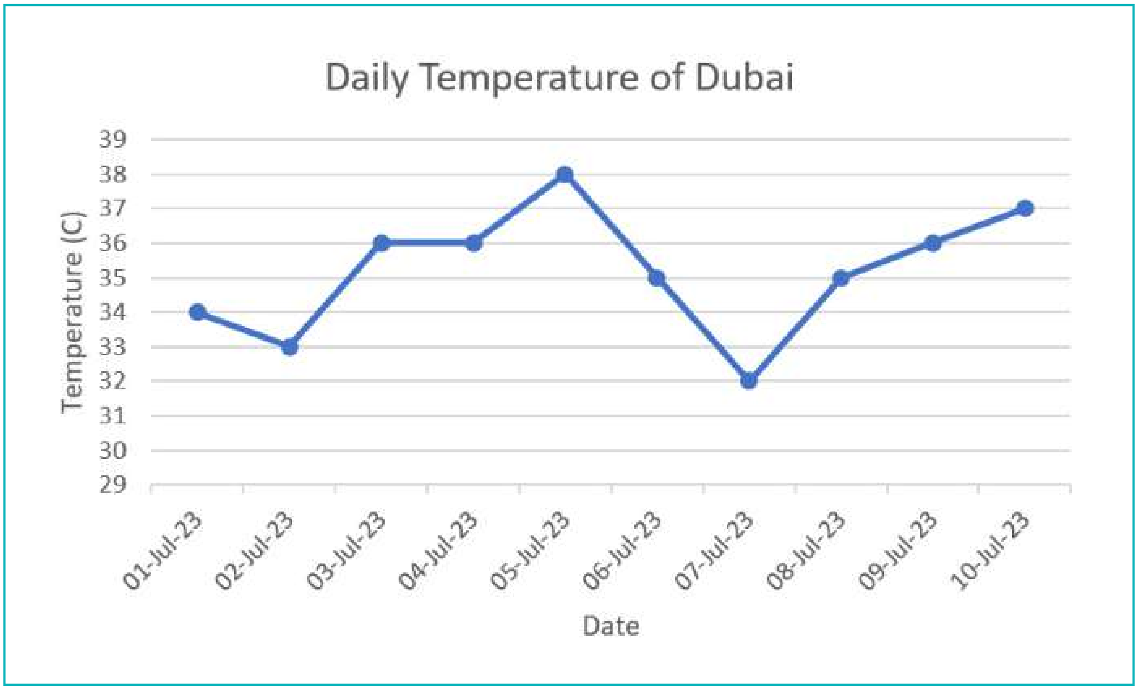
Steps

14 Add axis titles: Select **Axis Titles** in the **Quick Launch Options** as shown:

Graphs have an X- and a Y-axis, where we add the type of data that is displayed. The X-axis will show the dates and the Y-axis will show the temperature.



15 Change the Y-axis title to **Temperature (C)** and the X-axis title to **Date.**

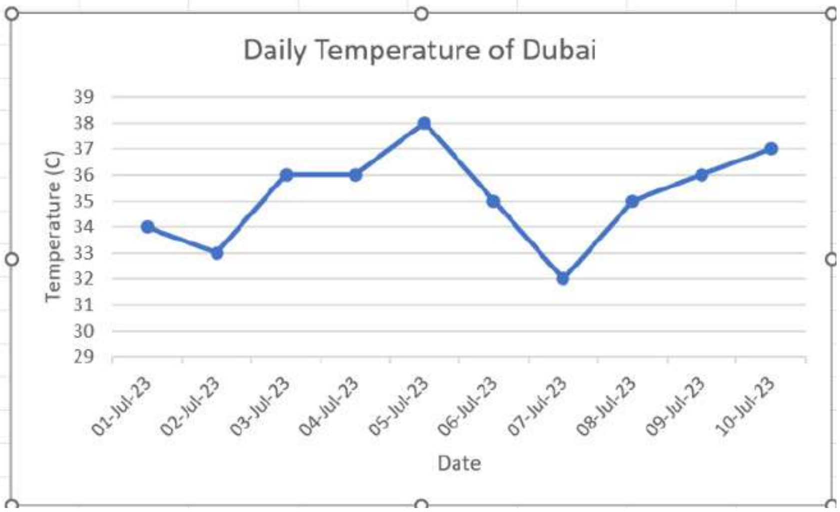


Steps

16 Press **Ctrl+S** to save the spreadsheet.

Final Output

Date	Temperature (C)
01-Jul-23	34
02-Jul-23	33
03-Jul-23	36
04-Jul-23	36
05-Jul-23	38
06-Jul-23	35
07-Jul-23	32
08-Jul-23	35
09-Jul-23	36
10-Jul-23	37

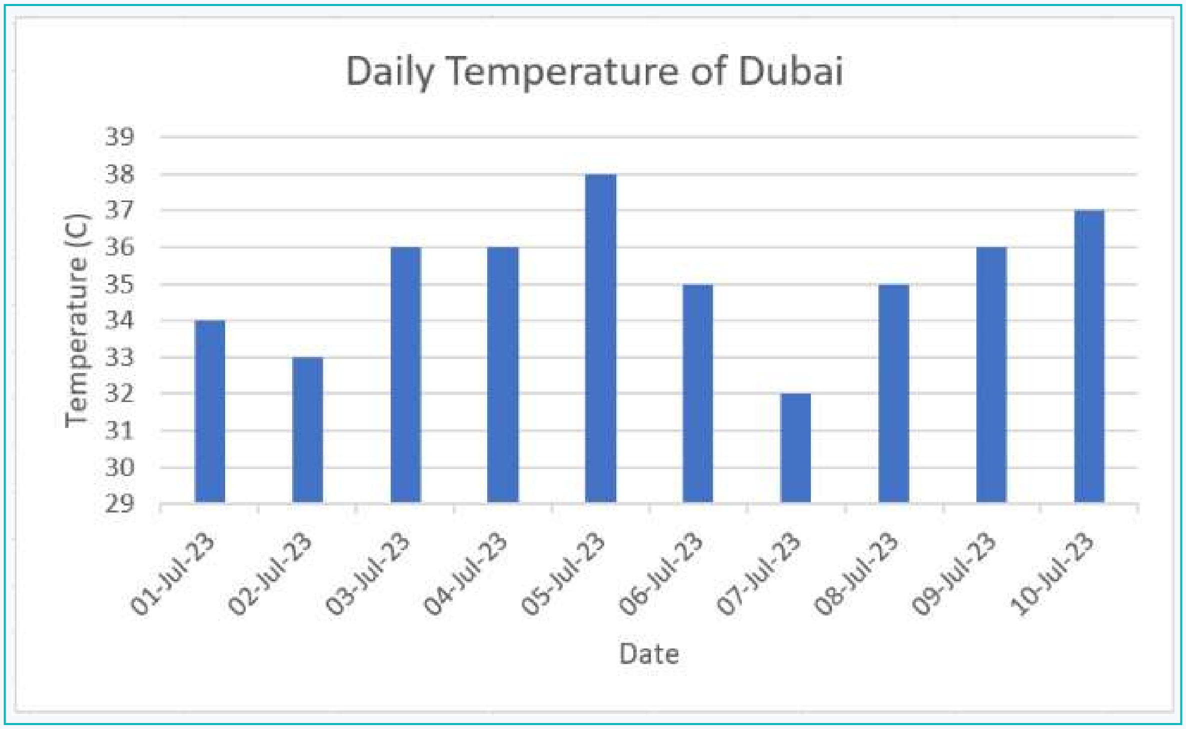


- In this graph, the **X-axis** (from left to right) shows the **dates** for the month of July. The **Y-axis** (from top to bottom) shows the **Temperature.**
- Each dot shows the temperature recorded each day.
- The line shows the variation in the temperature between 10 days.

Activity 2

To find out the date on which Dubai had the highest temperature, use the same data to create a bar graph.

Final Output



- We can look at the graph and tell that 5th July 2023 had the highest temperature.

A graph is an effective way to display data with lines, shapes, and colours. Graphs can be used to compare quantities or values. Graphs are useful because they are easier to understand than numbers and words alone. To interpret a graph, read the chart title, axis titles, and the entries. Then study the graph to understand what it shows. There are different types of graphs. Line graphs, bar graphs, and pie charts are a few different types of graphs.

10. Create Graphs

Tech Flash

Graphs : A graph is a pictorial representation that shows data with lines, bars, dots, shapes, or portions of a circle.

Line Graph : A line graph displays data that changes continuously over time.

Bar Graph : A bar graph uses bars that represent numbers to show data.

Chart : A chart represents data in the form of a table, graph, or diagram.

Pie Chart : A pie chart uses "pie slices" to show a portion of a part out of the total.